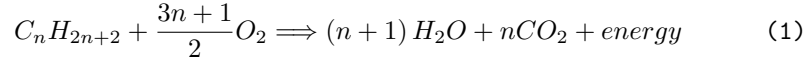


Bell Nozzle Rocket Engine Equation Database

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Propellant Stoichiometry of Hydrocarbon-Oxygen Propellant [1]

$$\dot{m} = \frac{m}{t} \quad (2)$$

Mass Flow Rate

$$\frac{1}{\gamma - 1} = \sum \frac{\text{mol}\%}{\gamma_i - 1} \quad (3)$$

Heat Capacity Ratio of a Multi-Compound Gas [2]

$$\frac{P}{P_s} = \left(1 + \frac{\gamma - 1}{2}Ma^2\right)^{-\frac{\gamma}{\gamma - 1}} \quad (4)$$

Isentropic Pressure Relation [3]

$$\frac{T}{T_s} = \left(1 + \frac{\gamma - 1}{2}Ma^2\right)^{-1} \quad (5)$$

Isentropic Temperature Relation [3]

$$\frac{\rho}{\rho_s} = \left(1 + \frac{\gamma - 1}{2}Ma^2\right)^{-\frac{1}{\gamma - 1}} \quad (6)$$

Isentropic Density Relation [3]

$$P = \rho R_s T \quad (7)$$

Ideal Gas Law

$$Ma = \frac{v}{c} \quad (8)$$

Mach Velocity [3]

$$c = \sqrt{\gamma \frac{P}{\rho}} = \sqrt{\gamma R_s T} \quad (9)$$

Speed of Sound [3]

$$q = \frac{1}{2}\rho v^2 = \frac{\gamma}{2}PMa^2 \quad (10)$$

Dynamic Pressure [3]

$$\epsilon = \left(\frac{\gamma + 1}{2}\right)^{\frac{1}{\gamma - 1}} \left(\frac{P_e}{P_c}\right)^{\frac{1}{\gamma}} \sqrt{\left(\frac{\gamma + 1}{\gamma - 1}\right) \left(1 - \left(\frac{P_e}{P_c}\right)^{\frac{\gamma - 1}{\gamma}}\right)} \quad (11)$$

Area Ratio [3]

$$P_c = \frac{\dot{m}\sqrt{T_t}}{A_t} \sqrt{\frac{R_s}{\gamma}} \left(\frac{\gamma + 1}{2} \right)^{\frac{\gamma+1}{2(\gamma-1)}} \quad (12)$$

Chamber Pressure [4]

$$A_t = \frac{\dot{m}}{\rho_t v_t} \quad (13)$$

Throat Area [5]

$$Ma_e = \sqrt{\frac{2}{\gamma - 1} \left(\left(\frac{P_c}{P_e} \right)^{\frac{\gamma-1}{\gamma}} - 1 \right)} \quad (14)$$

Isentropic Exit Mach Number

References

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- [2] O. Lanzi, "How do you calculate the heat capacity ratio for a multi-compound gas?." <https://chemistry.stackexchange.com/a/166796>, 2022.
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- [5] B. Brady, "Cryo rocket." <https://www.cryo-rocket.com>, 2020.